

Matrix Theory

Required Problems

Please submit your solutions to these problems to be graded. All solutions should be clearly written and include full explanation/justification.

1. (NS 2.5.5) Consider the standard basis E , and the basis $B = \{(1, 2, 3), (3, 2, 1), (0, 0, 1)\}$ for $\mathbb{R}^3$.
 - (a) Compute the change of basis matrices $[I]_{B,E}$ and $[I]_{E,B}$.
 - (b) Determine the matrices $[T]_{B,B}$, $[T]_{B,E}$, $[T]_{E,B}$, and $[T]_{E,E}$ for the linear operator T on $\mathbb{R}^3$ given by $T(a, b, c) = (6a + b, a - b - c, 2a - b + 2c)$.
2. Let V be the vector space of all differentiable functions from $\mathbb{R}$ to $\mathbb{R}$. Let $B = \{e^{5t}, e^{5t} \cos t, e^{5t} \sin t\}$ and consider the subspace $W = \text{span}(B)$ of V .
 - (a) Let D be the differentiation operator (i.e., $D(f(t)) = f'(t)$) on W . Find the matrix representation $[D]_{B,B}$.
 - (b) Verify Theorem 2.41 (i.e., $[Df]_B = [D]_{B,B}[f]_B$) for the input function $f(t) = 3e^{5t} - e^{5t} \cos t + 2e^{5t} \sin t$.
3. (variation on NS 2.5.6) Let $B := \{u_1, \dots, u_n\}$ and $C := \{v_1, \dots, v_n\}$ be ordered bases for a vector space V . Let T be a linear operator on V defined by $Tu_1 = v_1, \dots, Tu_n = v_n$. Show that $[T]_{B,B} = [I]_{B,C}$.

(Note. There is a typo in the statement of this problem in the book.)
4. Let A and B be square matrices. Prove or disprove each of the following:
 - (a) If A is equivalent to B , then A^2 is equivalent to B^2 .
 - (b) If A is similar to B , then A^2 is similar to B^2 .



Extra Practice Problems

This is a list of problems I recommend if you would like further practice to help you master the material. You are not expected to solve all of these problems, nor are you required to submit them for grading.

- Section 2.4: 2, 6, 7, 8, 10, 11, 13, 16, 19, 20, 21

- Section 2.5: 1, 2, 3, 4, 8

- Section 2.7: 50

1. Let V and W be finite vector spaces with $\dim(V) = \dim(W)$ and let $T : V \rightarrow W$ be a linear transformation. Show that there exist ordered bases B for V and C for W such that $[T]_{C,B}$ is a diagonal matrix.

